Kazakh-British Technical University

Data collection and preparation

STUDENTS' INDEPENDENT STUDY (SIS) №1

Weather & Tourism Trends

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Almaty

1. Introduction

When people choose where to travel, weather is usually one of the first things they think about. That made me curious whether climate is also connected to how "touristic" a city looks from the outside. For this project I tried to answer a simple question: do cities with warmer and drier weather have more tourist attractions listed on Wikipedia?

To answer it I collected data from two different sources, cleaned both datasets, joined them into one table and then looked at the numbers and charts. I picked ten well-known tourist cities from different parts of the world and climates: Paris, London, Rome, Barcelona, Istanbul, Dubai, Bangkok, Tokyo, New York and Almaty. Almaty was added on purpose to see how a city from Kazakhstan compares with the big tourist centres.

2. Data sources

Weather (API). I used the Open-Meteo Historical Weather API (https://archive-api.open-meteo.com/v1/archive). It is free, does not need an API key and returns clean JSON, which made it a good choice for a first project with an API. For every city I requested two daily values for the whole year 2025: the mean air temperature (temperature_2m_mean) and the total precipitation (precipitation_sum).

Tourist attractions (web scraping). For tourism I used English Wikipedia category pages of the form Category:Tourist_attractions_in_<City>. Each of these pages contains two lists: subcategories (for example "Museums in Paris") and individual pages (for example a specific museum or park). I treated every item in these lists as one attraction, so the number of items shows how well a city's attractions are represented on Wikipedia.

3. Data collection

The API part was straightforward. Open-Meteo works with coordinates, not city names, so I stored the latitude and longitude of each city in a small DataFrame. Then I looped over the cities and sent one GET request per city with the requests library, passing the coordinates, the date range 2025-01-01 to 2025-12-31 and the list of daily variables as parameters. Every request returned status 200 and 365 days of data. I turned the "daily" part of each JSON response into a DataFrame, added the city name and combined everything into one table of 3,650 rows (10 cities × 365 days). I also added a one-second pause between requests so as not to overload the server.

Before scraping Wikipedia I checked its robots.txt file with Python's RobotFileParser. It confirmed that pages under /wiki/ are allowed for crawlers. Wikipedia can block requests without a proper User-Agent, so I sent a short descriptive User-Agent header with every request. For each city I built the category URL, downloaded the page with requests and parsed the HTML with BeautifulSoup. The lists I needed are inside two blocks with the ids mw-subcategories and mw-pages, so I found these blocks and saved the text of every <li> element together with the city name and the block type. One small detail: in Wikipedia the category for New York is called New_York_City, so I kept a separate wiki_name column for the URL. All ten pages returned status 200, and the raw scraped table had 595 rows.

4. Data cleaning

The weather data was already quite clean, but the column names from the API were long and not very readable, so I renamed time to date, temperature_2m_mean to temp_c and precipitation_sum to rain_mm. The date column came as text, so I converted it to the datetime type, which was needed later for grouping by month. I checked the data for missing values and for duplicated city–date pairs and

found none. I still kept the rules for handling them in the code: a missing temperature would be filled by interpolation between the neighbouring days, and missing precipitation would be treated as 0 mm.

The scraped data needed more work. The names of subcategories included counters added by Wikipedia, for example "Cemeteries in Paris (1 C, 26 P)" and some names had extra spaces. I removed these with regular expressions and pandas string methods, so that the names became clean, like "Cemeteries in Paris". After that I checked for duplicates inside each city and found 9 names that appeared more than once in the same city's lists. After removing them the table went from 595 to 586 rows. I also replaced the technical block ids with simple labels "page" and "subcategory".

5. Merging the data

The two datasets have different levels of detail: weather is daily, while attractions are a list of names. To join them I first summarised both to one row per city. From the weather data I calculated the average temperature for the year, the total amount of rain and the number of rainy days (days with at least 1 mm of precipitation). From the attractions data I simply counted the number of items per city. Then I merged both summaries with the original city list using a left join on the city column. A left join makes sure that every city stays in the table even if one of the sources had no data for it; in that case the number of attractions would be set to 0. In my final run all cities had data in both sources, so the final table has 10 rows and no missing values.

6. Results

Table 1 shows the final merged dataset, sorted by the number of attractions.

City	Avg temp, °C	Total rain, mm	Rainy days	Attractions
Paris	12.8	740.9	120	100
London	12.4	613.4	107	79
Rome	17.3	1024.3	105	73
Barcelona	17.0	912.7	108	73
Tokyo	16.9	1322.1	126	63
Bangkok	28.2	1675.3	207	58
Istanbul	15.8	631.7	92	53
Dubai	28.5	37.1	7	42
New York	12.5	1116.1	119	30
Almaty	11.5	345.8	53	15

Table 1. Climate in 2025 and number of Wikipedia attractions per city

The descriptive statistics (Table 2) show that the cities are very different in climate. The average temperature ranges from 11.5 °C in Almaty to 28.5 °C in Dubai, and the yearly rain ranges from only 37.1 mm in Dubai to 1,675.3 mm in Bangkok. The number of attractions also varies a lot, from 15 to 100, with an average of about 59.

Statistic	Avg temp, °C	Total rain, mm	Rainy days	Attractions
Mean	17.3	841.9	104.4	58.6

Std	6.2	476.0	51.4	24.9
Min	11.5	37.1	7	15
Median	16.4	826.8	107.5	60.5
Max	28.5	1675.3	207	100

Table 2. Descriptive statistics of the merged dataset

To check the main question I calculated the Pearson correlation between the climate variables and the number of attractions (Table 3).

Pair of variables	**Correlation (r)**
Average temperature – attractions	-0.06
Total rain – attractions	0.23
Rainy days – attractions	0.37
Total rain – rainy days	0.93

Table 3. Correlation with the number of attractions

The correlation between temperature and attractions is almost zero (-0.06), so in this sample warmer cities do not have more attractions. The relation with rain is weak and, surprisingly, positive: 0.23 for total rain and 0.37 for rainy days. In other words, the cities with more rain had slightly more attractions, which is the opposite of what I expected. The very strong correlation between total rain and rainy days (0.93) is logical, because both measure the same thing in different ways.

7. Visualizations

Figure 1 shows the number of attractions per city. Paris is clearly the leader with 100 items, followed by London (79), Rome and Barcelona (73 each). Almaty has the fewest with 15.

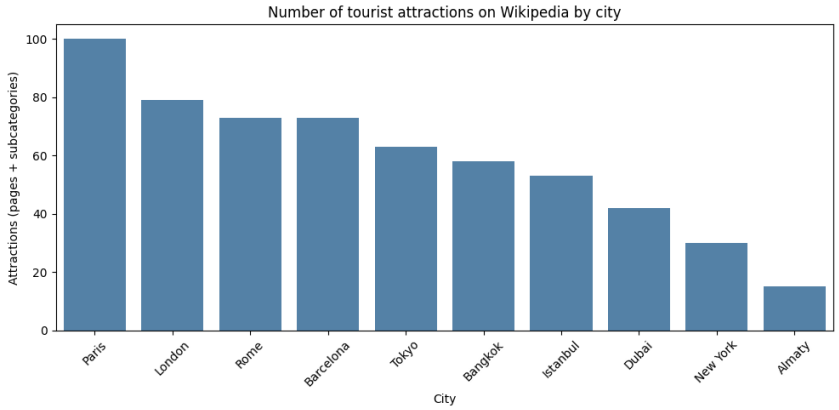


Figure 1. Number of tourist attractions on Wikipedia by city

Figure 2 puts temperature and attractions on one chart. If warm weather really meant more attractions, the points would go up from left to right. Instead they are spread out: the two warmest cities, Dubai and Bangkok, are on the right side but in the middle by height, while the cooler European cities are at the top.

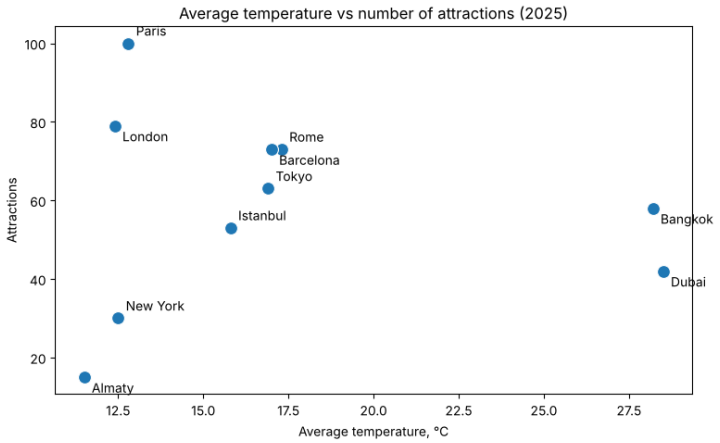


Figure 2. Average temperature vs number of attractions (2025)

Figure 3 shows the average temperature for every month. It explains why a single yearly average does not tell the whole story. Bangkok stays between 26 and 30 °C all year, while Almaty goes from -5 °C in January to 28 °C in July, a difference of more than 30 degrees. New York has a similar pattern, from about -2 °C in winter to 27 °C in summer. Dubai is the hottest city in summer, reaching 36 °C in August.

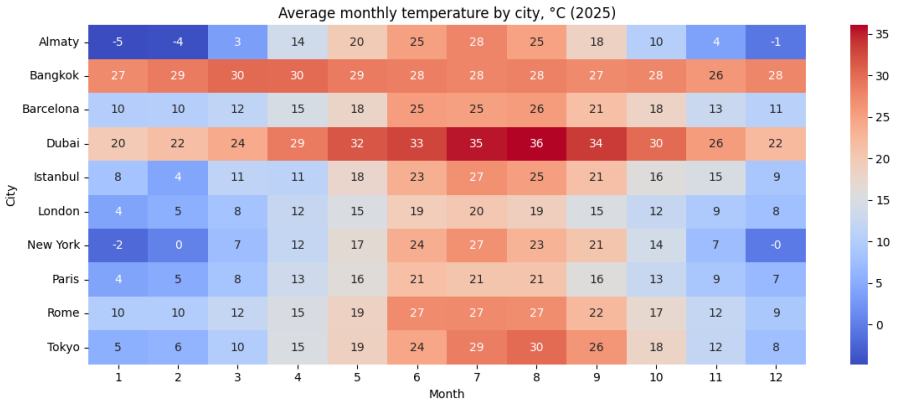


Figure 3. Average monthly temperature by city, °C (2025)

Figure 4 shows all correlations at once as a heatmap. The cells connected to attractions are pale, which visually confirms that the relations are weak.

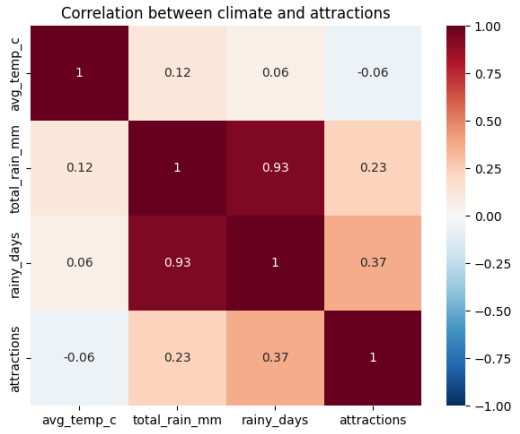


Figure 4. Correlation matrix of climate variables and attractions

8. Key insights

The main finding is that climate does not explain the number of tourist attractions in my data. Temperature has practically no relation with attractions, and rain has only a weak one, in the opposite

direction from what I expected. The top of the ranking is taken by European cities with a mild and fairly rainy climate, which suggests that history, culture and the age of a city matter much more than the weather.

Another insight is about the data source itself. The number of items on a Wikipedia category page shows how well a city is described on English Wikipedia, not how many tourists actually visit it. New York is one of the most visited cities in the world but has only 30 items here, while Paris has 100. Almaty having only 15 items probably reflects that it is less covered in English sources rather than that it has nothing to see.

9. Limitations and possible improvements

The sample is small: only 10 cities and one year of weather, so the correlations are not statistically reliable and can change a lot if a few cities are added. A subcategory such as "Museums in Paris" is counted as one attraction, even though it may contain dozens of pages. In a future version I would follow the subcategories to count all pages inside them, use more cities, add a real tourism measure such as the number of international visitors, and compare the weather during the main tourist season instead of the yearly average.

10. Conclusion

In this project I built the full pipeline from the course: collecting data from an API and from a web page, cleaning it, merging two different sources, calculating statistics and making charts. Technically everything worked: all 10 API requests and all 10 Wikipedia pages were collected, and the final dataset has no missing values or duplicates. The answer to my question turned out to be negative: in these ten cities, a warmer or drier climate is not connected with more tourist attractions on Wikipedia.